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AMRITA SCHOOL OF ENGINEERING, BENGALURU

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

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ASSIGNMENT FOR 23ECE302 VLSI DESIGN

SYSTEM DESIGN FOR 23ECE383 VLSI DESIGN LABORATORY

REPORT

JULY TO NOVEMBER 2026

Implementation of a Compact Multiphase Divide-by-Two CMOS Frequency Divider in 45 nm CMOS (Cadence GPDK045)

Submitted by

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TITLE OF THE SYSTEM DESIGN PROJECT:

Implementation of a Compact Multiphase Divide-by-Two CMOS Frequency Divider in 45 nm CMOS (Cadence GPDK045)

TITLE OF THE PAPER:

A Compact Multiphase Divide-by-Two CMOS Frequency Divider

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CIRCUIT DIAGRAM:

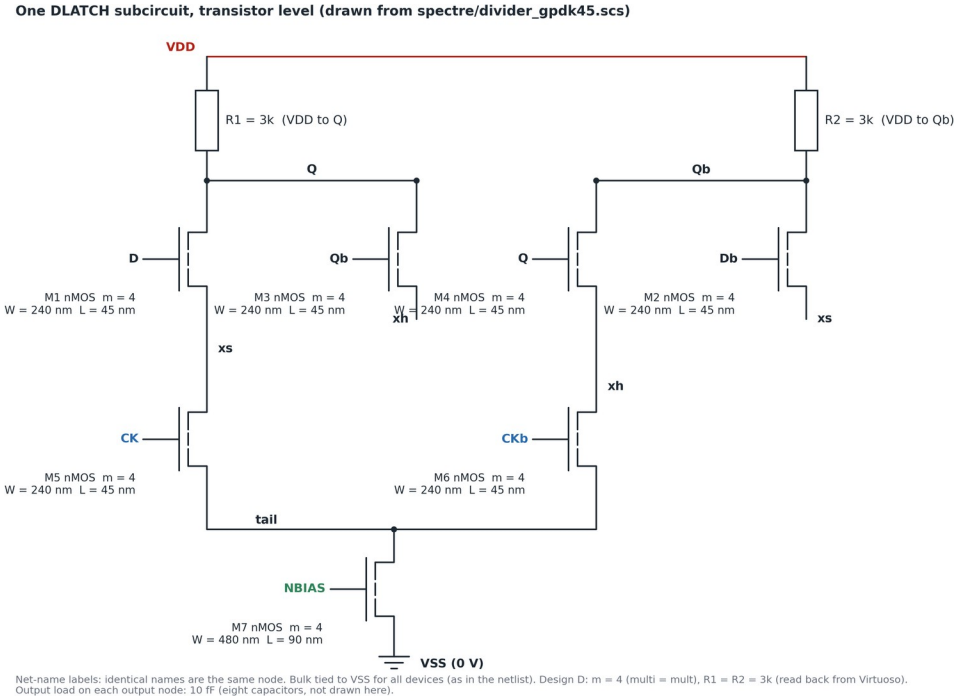


Figure 1: Schematic of the differential DLATCH stage (M1-M7, R1, R2). Sizing: M1-M6 W = 240 nm, L = 45 nm; M7 W = 480 nm, L = 90 nm. Values shown for design D (m = 4, R1 = R2 = 3 kOhm; baseline uses m = 1, R = 8 kOhm).

THEORY AND WORKING PRINCIPLE:

Topology. The divider consists of four differential D-latches in a ring with alternating cross-connections (Q1b-D2, Q2-D3, Q3b-D4, Q4-D1b) and inverted clocks on latches 2 and 3 (Kuo et al.). Each latch contains sampling pair M1-M2, clock switches M5-M6, regenerative hold pair M3-M4, tail transistor M7, and loads R1-R2. Sampling occurs when CK is high, and regeneration occurs when CK is low, producing eight phases at half the input frequency.

Elmore delay model. Latch propagation delay is modeled as t_pd = (8C)(R/2) = 4RC, where each output carries effective capacitance 8C and load R/2 = R1. Division requires T_CK/2 > 2 Td. For load capacitance C_L = 10 fF, the parasitic-free lower bound is Td >= R1 C_L, yielding 30 ps for design D (R1 = 3 kOhm) and 20 ps for design B (R1 = 2 kOhm). At 1.4 GHz (design D), 2 Td >= 60 ps satisfies T_CK/2 = 357 ps. At 6 GHz (design B), 2 Td >= 40 ps satisfies T_CK/2 = 83 ps. Design B divides up to 6 GHz and fails at 6.2 GHz (T_CK/2 = 80.6 ps). Its measured CK-to-Q1 delay of 66.10 ps is 3.3x the intrinsic lower bound (79% of T_CK/2), showing that parasitic capacitances dominate near the frequency limit.

Stack resistance and regeneration. The pull-down network forms a three-transistor stack (M1, M5, M7) with effective resistance 3 Ron/m. Latch regeneration requires gm R1 > 1, yielding R1 >= 1.1 x (3 Ron/m). With Ron = 2.3 kOhm, the unit stack is 6.90 kOhm, requiring R1 >= 7.59 kOhm. The paper load (1.15 kOhm) falls far below this threshold, preventing division. Measured thresholds for m = 1 span 6 to 7 kOhm at vb = 1 V, and 7 to 8 kOhm at vb = 0.8 V. For m = 4, the required threshold is 1.90 kOhm (band 1.88 to 2.25 kOhm); design D (R1 = 3 kOhm) provides margin, while design B (R1 = 2 kOhm) operates within the marginal band. Measured division agrees with the rule for 5 of 5 tested parameter pairs. The unit device aspect ratio W/L = 5.33 (240 nm / 45 nm) matches the reference design ratio of 5.00 (0.9 um / 180 nm).

Phase-step relationship. Outputs toggle in pairs separated by t_pair (measured between rising edges of Q2 and Q3b), yielding phase step a = 180 t_pair f_in degrees, alternating with 90 - a degrees. For design D at 1.4 GHz, t_pair = 25.95 ps gives steps of 83.46 and 6.54 deg (a = 180 x 25.95 ps x 1.4 GHz = 6.54 deg). For design B at 6 GHz, t_pair = 38.16 ps gives 48.78 and 41.22 deg (a = 41.22 deg). The unsized baseline divider (m = 1, R = 8 kOhm) at 1.4 GHz gives 71.42 and 18.59 deg (t_pair = 73.75 ps, a = 18.59 deg). Nominal 45 deg spacing requires T_CK = 4 t_pair: 103.8 ps (9.63 GHz) for design D and 152.7 ps (6.55 GHz) for design B. Phase error decreases as frequency increases toward the divider limit, matching the reference paper value (3.75 deg) only in design B at 6 GHz.

Definitions of performance metrics. Delay is the average time from each CK rising edge to the next transition on any output (25.99 ps for design D; single-edge CK-to-Q1 delay is 51.95 ps). Power is VDD times the average supply current over full cycles. PDP is power times delay (fJ). Division is valid when all eight outputs toggle at f_in/2 within 0.1 percent with peak-to-peak swing >= 0.1 V.

TOOL & TECHNOLOGY USED:

Tools and environment: Cadence Virtuoso IC615 (6.1.5), Spectre circuit simulator (MMSIM 12.1), and Assura physical verification (4.1). Process technology is Cadence GPDK045 v6.0 (45 nm generic CMOS), using standard-threshold NMOS devices (g45n1svt, 1.0 V nominal) evaluated across tt, ff, ss, fs, and sf corners.

Device sizing: unit transistors M1-M6 have W = 240 nm, L = 45 nm, and tail transistor M7 has W = 480 nm, L = 90 nm. Device multiplier m scales channel widths in parallel.

Operating conditions: nominal supply VDD = 1 V (PVT range 0.9 V to 1.1 V), nominal temperature 27 C, clock transition times 20 ps, tail bias vb = 1 V, and load capacitance 10 fF per output. Simulations were conducted using 40 ns transient analysis runs in Cadence Spectre.

Virtuoso schematic parameter extraction for the design D DLATCH cell: R1 and R2 = 3 kOhm; M1-M6 w = 240n, l = 45n, m = 4; M7 w = 480n, l = 90n, m = 4. These match Table 2 and the nominal transistor dimensions.

Circuit schematics for the DLATCH and divider core are illustrated in Figure 1 and Figure 2 (ring architecture in Figure 8).

SCREENSHOT OF THE SIMULATOR BASED CIRCUIT IMPLEMENTATION:

Unsized baseline (m = 1, R = 8 kOhm): unit devices (m = 1), load R1 = R2 = 8 kOhm, tail bias vb = 1 V, and 1.4 GHz input under nominal conditions (tt corner, 27 C, VDD = 1 V). All eight outputs divide to 700.0 MHz, with Q1 swinging from 0.749 V to 0.978 V.

Selection of R: load resistance was swept from 5 to 12 kOhm (12 runs) at vb = 0.8 V and 1 V. Division fails up to 6 kOhm and succeeds from 7 kOhm at vb = 1 V; at vb = 0.8 V it fails up to 7 kOhm and divides from 8 kOhm. The value R = 8 kOhm at vb = 1 V was selected as the minimum round value dividing at both biases.

Failure of paper sizing: direct implementation of the paper rule (m = 1, R1 = R2 = Ron/2 = 1.15 kOhm, Ron = 2.3 kOhm) fails to divide at 1.4 GHz. Output Q1 oscillates at 2799.8 MHz (2.0 x f_in) with only 15.3 mV swing (Figure 9). Supply power is 0.165 mW; delay, PDP, and phase error are undefined.

Mechanism: the pull-down path comprises three stacked devices (M1, M5, M7) with effective resistance 3 Ron = 6.90 kOhm for m = 1. Regeneration requires R1 >= 1.1 x 3 Ron = 7.59 kOhm, consistent with simulated bounds of 6-7 kOhm at vb = 1 V and 7-8 kOhm at vb = 0.8 V.

Table 1: Device list, circuit without the sizing concept

Device	W	L	Multiplier	Role
M1, M2	240 nm	45 nm	1	sampling pair (differential inputs)
M3, M4	240 nm	45 nm	1	hold pair (cross-coupled regenerative pair)
M5	240 nm	45 nm	1	switch controlled by CK
M6	240 nm	45 nm	1	switch controlled by CKb
M7	480 nm	90 nm	1	tail current source, gate at vb = 1 V
R1, R2	n/a	n/a	n/a	load resistors, 8 kOhm each

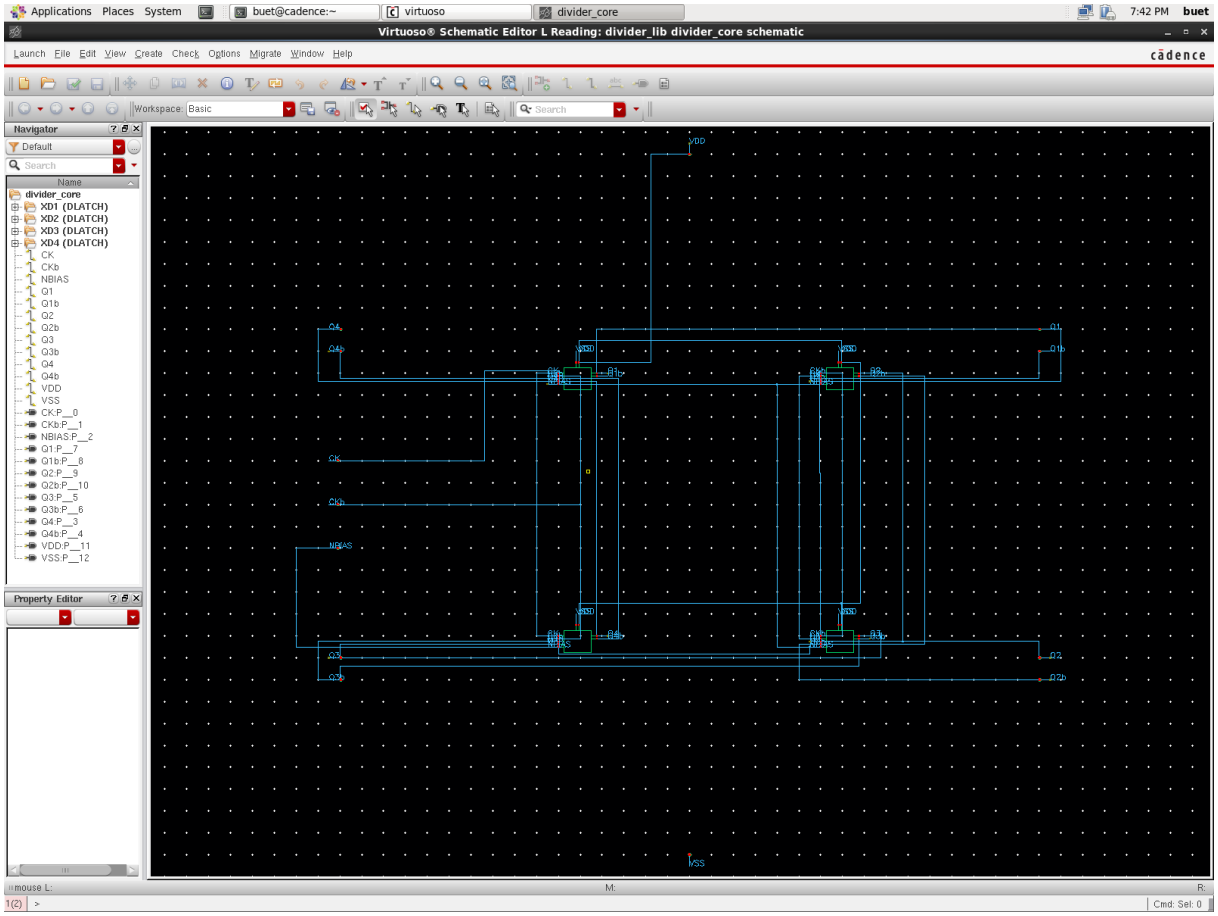


Figure 2: Virtuoso schematic editor window showing the divider core with four DLATCH instances (XD1 to XD4) connected in a ring. Baseline and design D share this topology, differing only in m and R (Table 1 and Table 2). Schematic checks: 0 errors and 23 warnings for DLATCH; 0 errors and 11 warnings for divider_core; 0 errors and 19 warnings for tb_divider. The warnings are floating-net notices at net-label stubs; the netlist is unchanged by the wiring.

SCREENSHOT OF THE OUTPUT WAVEFORM OBTAINED FROM THE SIMULATOR (DC AND TRANSIENT):

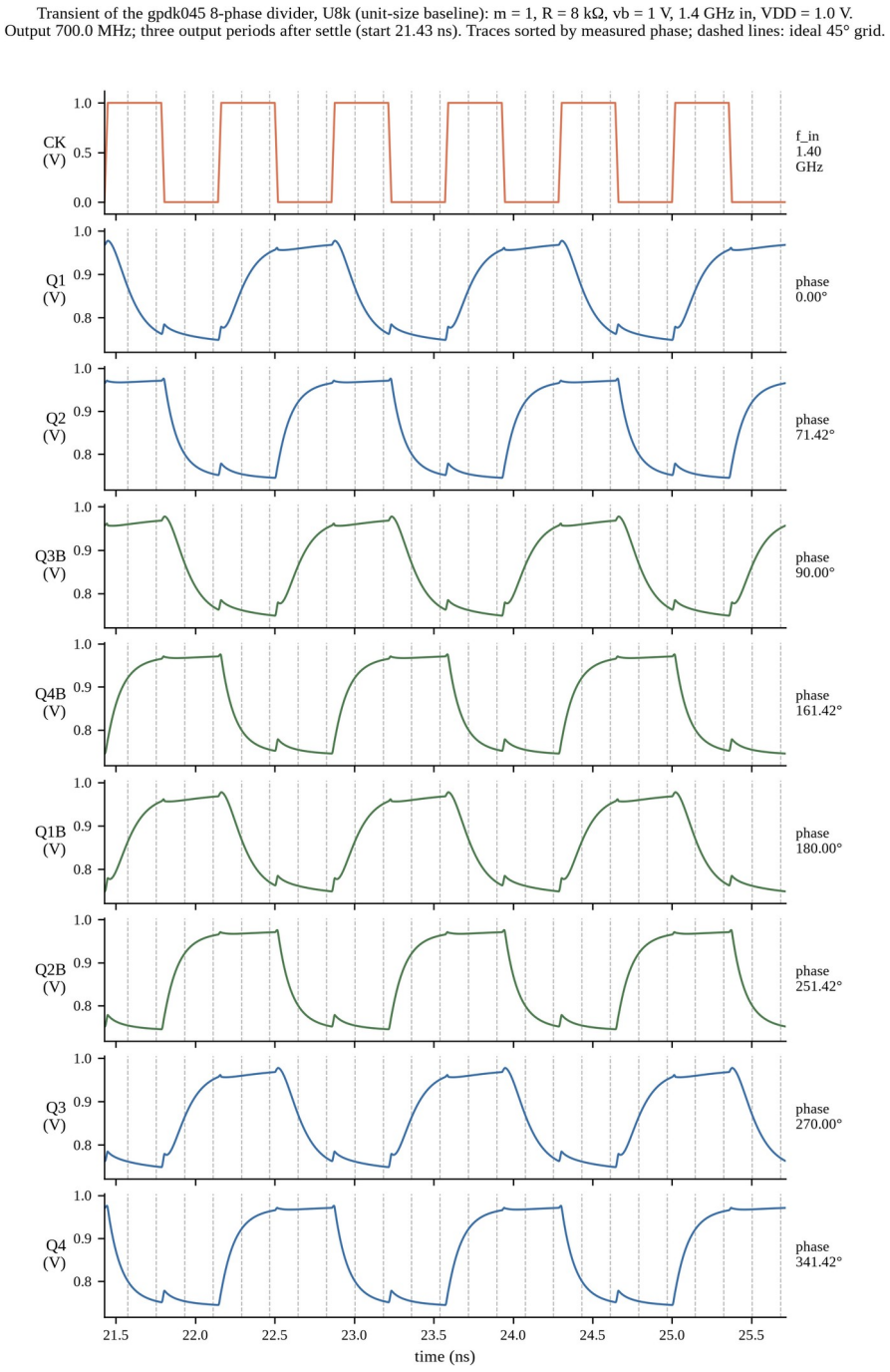


Figure 3: Transient waveforms for the baseline divider without sizing (m = 1, R1 = R2 = 8 kOhm, vb = 1 V) at 1.4 GHz. All eight outputs divide to 700.0 MHz, with Q1 swinging from 0.749 V to 0.978 V. Waveforms for paper sizing appear in Figure 9.

Simulation driven directly from the Virtuoso schematic (OCEAN)

The testbench schematic of the sized divider (design D: R1 = R2 = 3 kOhm, m = 4) was simulated with Spectre directly from Virtuoso at the typical corner (tt, 27 C, VDD = 1.0 V, bias 1.0 V) with a 1.4 GHz input clock. All eight outputs divide to 700 MHz (ratio 2.0). The clock-to-output delay is 26.66 ps and the average supply power is 0.455 mW. Figure 5 shows the corresponding ViVA waveforms.

The standalone netlist of the same circuit gives 0.541 mW and 25.99 ps. The difference comes from the layout-dependent device parameters (stress, well-proximity and series-resistance terms) that the Virtuoso netlister adds to every transistor. Adding the same parameters to the standalone netlist gives 0.455 mW and 26.66 ps, identical to the schematic-driven run. With these parameters the power-delay product of design D is 12.14 fJ instead of 14.07 fJ, and the division is unchanged. The tables in this report use the standalone netlist, which has the default device parameters of the process.

SCREENSHOT OF THE CIRCUIT BASED ON SIZING CONCEPT:

Sized divider (design D): device multiplier m = 4, load R1 = R2 = 3 kOhm, and input 1.4 GHz. The pull-down stack resistance is 3 Ron/m = 1.72 kOhm, requiring load resistance of 1.90 kOhm (tolerance band 1.88 to 2.25 kOhm). Load R = 3 kOhm provides adequate margin. The same sizing rule applies to design B at 6 GHz (m = 4, R1 = R2 = 2 kOhm). Transistor M7 operates in triode, with transient drain voltage remaining below Vdsat.

Comparison with baseline (Table 3): sizing cuts delay from 62.21 ps (unsized baseline) to 25.99 ps (design D) (2.39x) and increases swing from 0.2289 V to 0.4344 V. Power rises from 0.1375 mW to 0.5413 mW (3.94x), PDP from 8.557 fJ to 14.067 fJ (1.64x), and phase-step error from 26.42 deg to 38.46 deg at 1.4 GHz.

Table 2: Device list, circuit based on the sizing concept

Device	W	L	Multiplier	Role
M1, M2	240 nm	45 nm	4	sampling pair (differential inputs)
M3, M4	240 nm	45 nm	4	hold pair (cross-coupled regenerative pair)
M5	240 nm	45 nm	4	switch controlled by CK
M6	240 nm	45 nm	4	switch controlled by CKb
M7	480 nm	90 nm	4	tail current source, gate at vb = 1 V
R1, R2	n/a	n/a	n/a	load resistors, 3 kOhm each

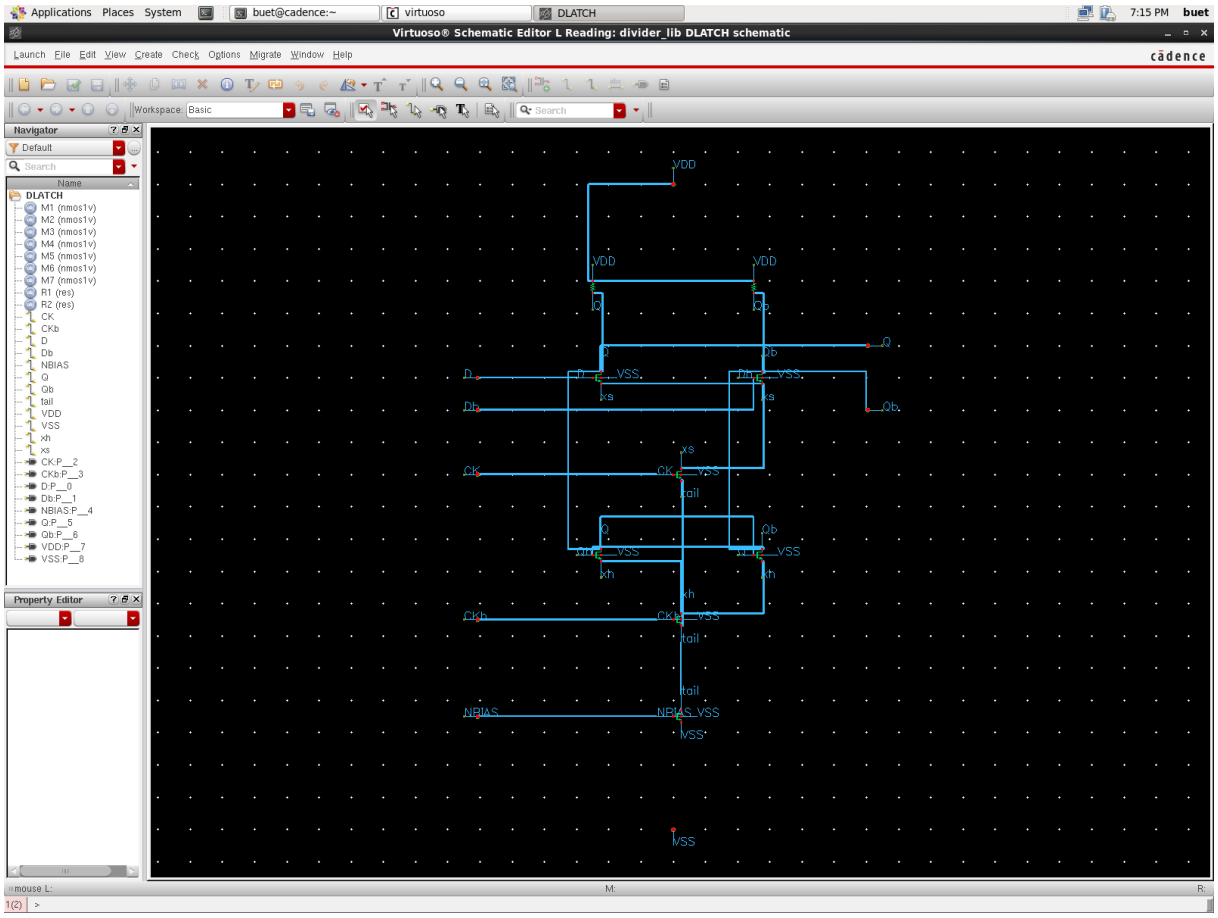


Figure 4: Virtuoso schematic editor window showing the DLATCH cell with sizing applied (design D: m = 4, R1 = R2 = 3 kOhm), consistent with Table 2. Schematic checks: 0 errors and 23 warnings for DLATCH; 0 errors and 11 warnings for divider_core; 0 errors and 19 warnings for tb_divider. The warnings are floating-net notices at net-label stubs; the netlist is unchanged by the wiring.

SCREENSHOT OF THE OUTPUT WAVEFORM OBTAINED FROM THE SIMULATOR FOR THE CIRCUIT WITH SIZING CONCEPT (DC AND TRANSIENT):

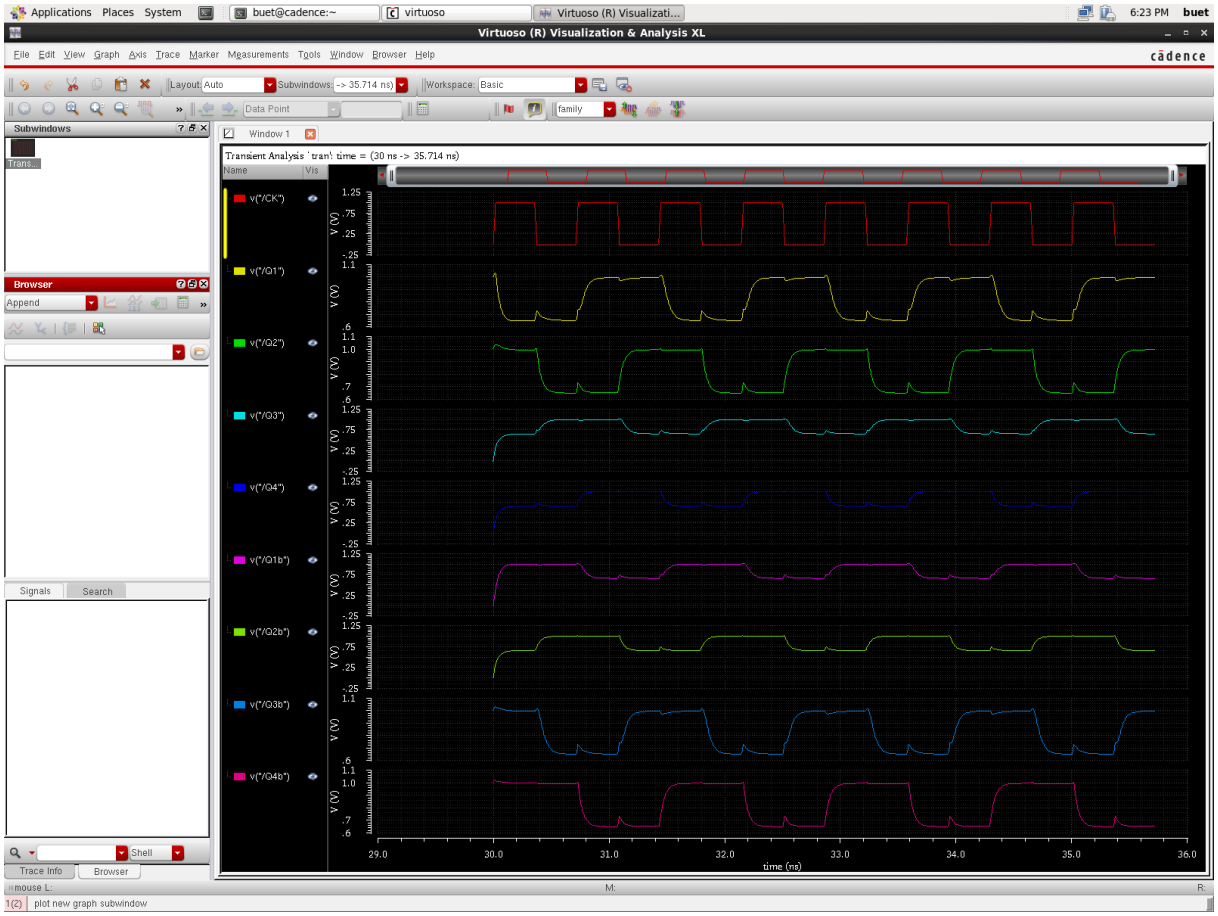


Figure 5: ViVA transient waveforms of design D from the Virtuoso testbench via Spectre.

PROVIDE THE VALUES FOR THE FOLLOWING ALONG WITH THE SCREENSHOTS AS PROOF:

Table 3: Performance values with and without the sizing concept, with the proof figures for each row

Parameter	Value without Sizing	Value with Sizing	Proof (figure)
Power calculation (P)	0.1375 mW	0.5413 mW	Fig. 15 (supply current, both columns)
Delay calculation (t_pd)	62.21 ps	25.99 ps	Calculated mean delay (see note)
Power Delay Product (PDP)	8.557 fJ	14.067 fJ	Calculated: power x delay
Input frequency (f_in)	1.4 GHz	1.4 GHz	Fig. 3 (without), Fig. 5 (with)
Output frequency (f_out)	700.0 MHz	700.0 MHz	Fig. 3 (without), Fig. 5 (with)
Max phase-step error	26.42 deg	38.46 deg	Fig. 11
Output swing	0.2289 V	0.4344 V	Fig. 3 (without), Fig. 5 (with)
Tail-device saturation check (M7)	**Vds = 0.0671 V, Vdsat = 0.3578 V: NOT saturated (triode)**	**Vds = 0.0669 V, Vdsat = 0.3578 V: NOT saturated (triode)**	Fig. 12 (without), Fig. 13 (with)

Notes: Baseline without sizing corresponds to unit devices (unsized baseline: m = 1, R = 8 kOhm, vb = 1 V). The paper sizing circuit (m = 1, R = 1.15 kOhm) fails to divide (supply power 0.165 mW; delay, PDP, and phase error undefined; Figure 9). Delay is the average time from CK rising edges to the next output transition (first-edge CK-to-Q1 delay: 135.96 ps for unsized baseline, 51.95 ps for design D). Vds indicates peak transient tail voltage V(XD1.tail); Vdsat is the extracted saturation voltage for M7.

PDP is computed from unrounded power and delay values.

SCREENSHOT OF THE PVT ANALYSIS:

Table 4: PVT summary for design D at 1.4 GHz (41 of 45 corners pass)

Corner	Temp (°C)	0.9 V	1.0 V	1.1 V
tt	-40	PASS	PASS	PASS
tt	27	PASS	PASS	PASS
tt	125	**FAIL**	PASS	PASS
ff	-40	PASS	PASS	PASS
ff	27	PASS	PASS	PASS
ff	125	PASS	PASS	PASS
ss	-40	PASS	PASS	PASS
ss	27	PASS	PASS	PASS
ss	125	**FAIL**	PASS	PASS
fs	-40	PASS	PASS	PASS
fs	27	PASS	PASS	PASS
fs	125	**FAIL**	PASS	PASS
sf	-40	PASS	PASS	PASS
sf	27	PASS	PASS	PASS
sf	125	**FAIL**	PASS	PASS

Table 5: PVT summary for design B at 6.0 GHz (21 of 45 corners pass)

Corner	Temp (°C)	0.9 V	1.0 V	1.1 V
tt	-40	PASS	PASS	PASS
tt	27	**FAIL**	PASS	PASS
tt	125	**FAIL**	**FAIL**	**FAIL**
ff	-40	PASS	PASS	PASS
ff	27	**FAIL**	PASS	PASS
ff	125	**FAIL**	**FAIL**	**FAIL**
ss	-40	**FAIL**	PASS	PASS
ss	27	**FAIL**	**FAIL**	PASS
ss	125	**FAIL**	**FAIL**	**FAIL**
fs	-40	PASS	PASS	PASS
fs	27	**FAIL**	PASS	PASS
fs	125	**FAIL**	**FAIL**	**FAIL**
sf	-40	**FAIL**	PASS	PASS
sf	27	**FAIL**	**FAIL**	PASS
sf	125	**FAIL**	**FAIL**	**FAIL**

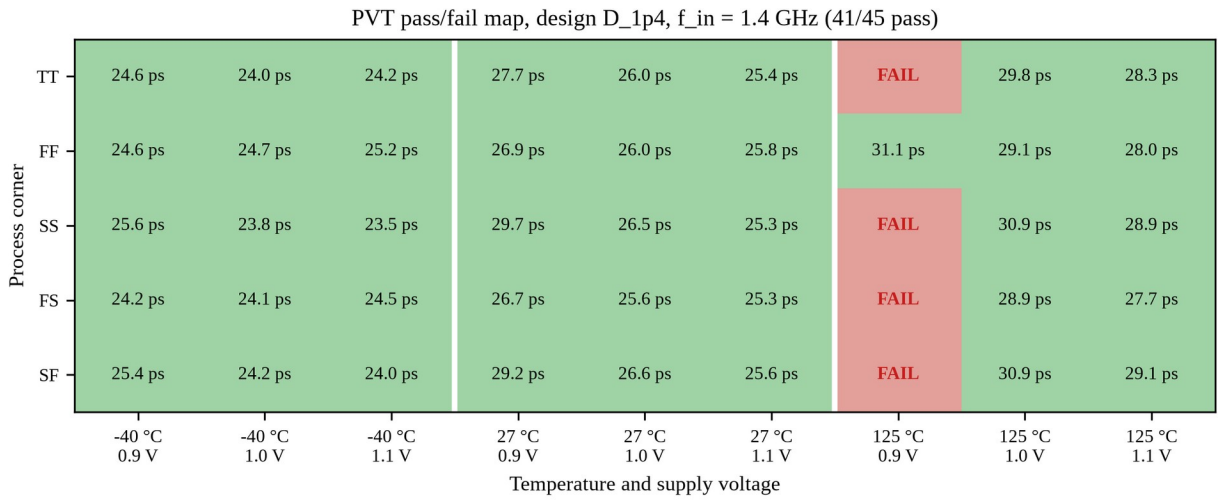


Figure 6: PVT pass/fail map for design D at 1.4 GHz across 5 corners, temperatures -40/27/125 C, and supplies 0.9/1/1.1 V (41 of 45 points pass).

SCREENSHOT OF THE LAYOUT:

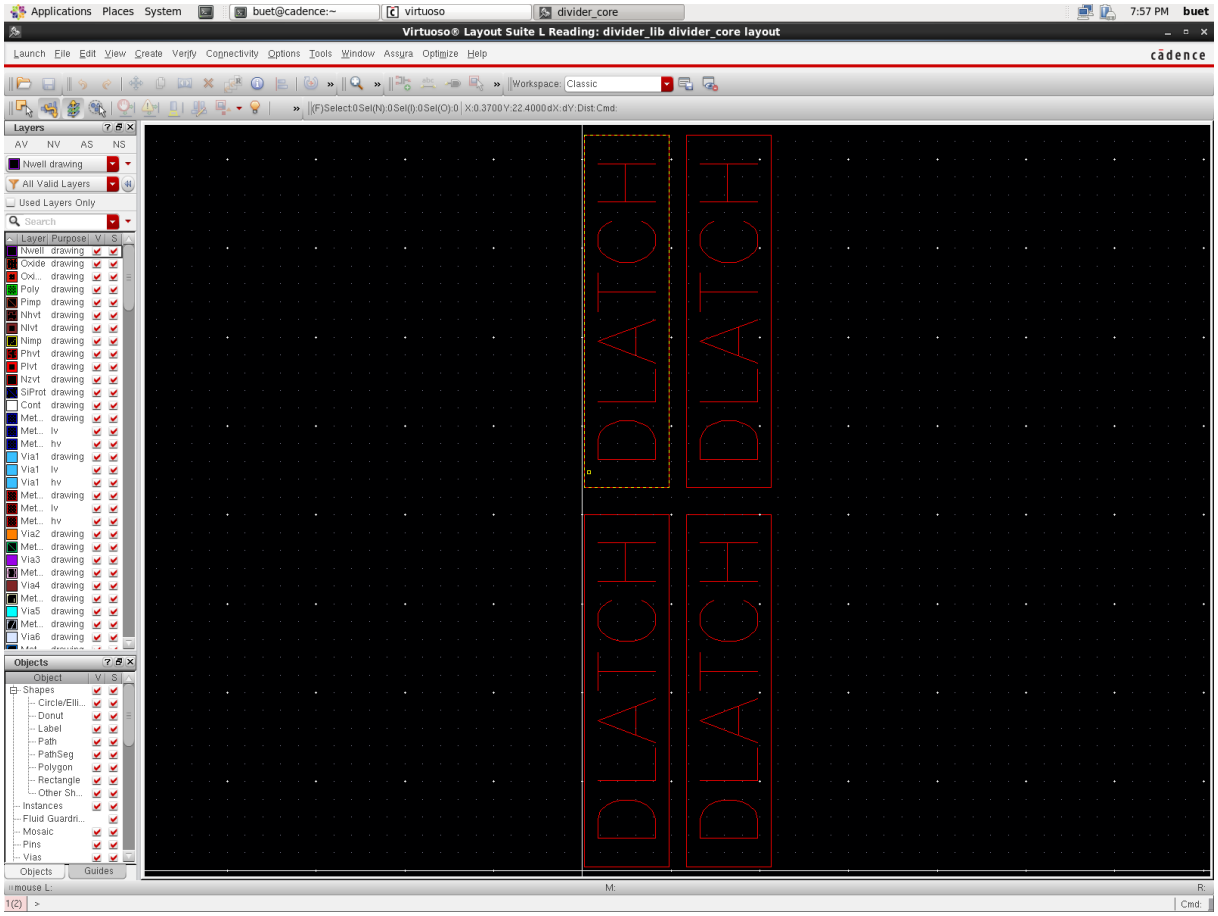


Figure 7: Virtuoso layout editor showing divider_core with four DLATCH instances (XD1 to XD4) placed in a 2x2 array (placement only, unrouted).

SCREENSHOT OF THE OUTPUT WAVEFORM OBTAINED FROM THE SIMULATOR FOR THE LAYOUT:

The layout is placement only: it is not routed, and LVS and post-layout simulation were not run.

ANY OTHER RELEVANT INFORMATION:

Layout. The DLATCH cell contains 7 unit nMOS pcells (0.595 um wide) and 2 poly resistors (segment width 0.8 um, segment length 7.855 um, 200 ohm/sq, target resistance 2000 ohm). Transistors M1-M6 use 2 fingers of 120 nm (240 nm, L = 45 nm); M7 uses 4 fingers of 120 nm (480 nm, L = 90 nm) for m = 1. Design D requires 4x width (m = 4) and 3 kOhm load (baseline uses 8 kOhm). DLATCH boundary is 4.875 x 20.28 um. Cell divider_core places XD1 R0, XD2 MY, XD3 MX, XD4 R180 with a 1 um gap (clock-star length 98.62 um each). An automatic routing run with Cadence Chip Assembly Router completed 136 of 137 connections but produced 134 DRC errors and mismatches in LVS, so the placement layout is reported. Cell layouts are shown in Figure 7 (divider_core) and Figure 21 (DLATCH).

DRC verification. Assura DRC on DLATCH reports 21 errors across 3 rules (Figure 22). Rules 220 (1) and 221 (6) give 7 substrate-stamp errors from unrouted taps at device pitch 0.755 um. Rule 595 (14) gives 14 Metal1 minimum-area errors (threshold 0.02 um^2); 4 markers measure 0.06 x 0.12 um (0.0072 um^2) across x = 0.96-1.155 um, y = 1-2.02 um. Contact pins (0.2 x 0.15 um = 0.03 um^2) meet the minimum area. The unrouted layout is not DRC clean.

Schematic connectivity. Pin-by-pin comparison of the 4 DLATCH instances (XD1-XD4) against the netlist confirmed full consistency with 0 mismatches. Schematic checks: 0 errors and 23 warnings for DLATCH; 0 errors and 11 warnings for divider_core; 0 errors and 19 warnings for tb_divider. The warnings are floating-net notices at net-label stubs; the netlist is unchanged by the wiring.

Design B operating point. At 6 GHz (m = 4, R1 = R2 = 2 kOhm), the divider achieves 3000.0 MHz output, 0.4929 mW power, 27.94 ps delay, 13.768 fJ PDP, and 3.78 deg maximum phase error (48.78 and 41.22 deg steps). Output Q1 swings 0.135 V (0.818 V to 0.953 V), above the 0.1 V limit. Peak tail voltage reaches 66.2 mV, below Vdsat = 357.8 mV (triode, Figure 14). PVT results appear in Table 5.

DC operating points. At t = 0 (clocks low, switches off), tail current is negligible (Vgs = 1 V). For paper sizing, M7 is in triode (Vds = 56.3 nV, Ids = 35.6 pA). For design D, M7 is in triode (Vds = 54.4 nV, Ids = 137.4 pA). For design B (m = 4, vb = 1 V), M7 is in triode (Vds = 54.4 nV, Ids = 137.4 pA).

Tail transistor operating region. With Vgs = 1 V and Vth = 527.5 mV, extracted Vdsat is 357.8 mV (at t = 0, gm = 200.9 pS, gds = 2.531 mS, Vds = 54 nV). In transient operation, V(XD1.tail) spans 8.1 to 66.9 mV for design D (XD2 reaches 68.2 mV) and 8.0 to 67.1 mV for the unsized baseline. Because Vds remains below Vdsat, M7 operates in triode rather than saturation (Figures 12 and 13).

Frequency sweeps. Design B (m = 4, R = 2 kOhm) was swept from 1.4 to 6.2 GHz: it divides at 1.4, 2, 3, 4, 5, 5.5, 5.8, 6 GHz and fails at 6.2 GHz. Design D (m = 4, R = 3 kOhm) was swept from 3 to 8 GHz: it divides at 3, 4, 5, 5.5 GHz and fails at 6, 6.5, 7, 8 GHz (the 1.4 GHz operating point also divides). Design E (m = 6, R = 2 kOhm) was swept from 3 to 8 GHz: it divides at 3, 4, 5, 5.5, 6, 6.5, 7 GHz and fails at 8 GHz (1.4 GHz divides at tt, 27 C, 1.0 V). Maximum phase-step error at the highest dividing frequency is 3.78 deg for design B, 13.82 deg for design D, and 15.24 deg for design E, compared with 38.68, 30.89, and 34.32 deg at the lower frequency bounds. In all cases, phase error improves as frequency increases (Figure 16).

Parameter exploration (16 points): m = 4, R = 2.2 kOhm divides up to 6.25 GHz (5.57 deg phase-step error there); m = 4, R = 2.5 kOhm divides up to 6 GHz (11.23 deg phase-step error there) and fails at 6.25 GHz; m = 5, R = 2.2 kOhm divides up to 6.25 GHz (16.56 deg phase-step error there); m = 5, R = 2.5 kOhm divides up to 6.25 GHz (14.36 deg phase-step error there). These nominal simulations characterize the upper operating boundary near the frequency limit of design B.

PVT analysis. Across PVT corners, design D at 1.4 GHz: 4 of 45 corners fail, all at 125 C and 0.9 V. At 1.4 GHz, design B: 18 of 45 corners fail (3 at 27 C and 0.9 V, all 15 at 125 C), and design E (m = 6, R = 2 kOhm): 4 of 45 corners fail, all at 125 C and 0.9 V. At 6 GHz, design B: 24 of 45 corners fail (2 at -40 C and 0.9 V, 7 at 27 C, all 15 at 125 C). Passing corners of design D span 0.246-1.078 mW and 23.54-31.05 ps. In Table 6 (Appendix), ratio is f_in/f_out (ratio 2 denotes correct division; 0.5 denotes outputs at 2 f_in).

Transient delay and edge times. First-edge CK-to-Q1 delays are 135.96 ps (unsized baseline), 51.95 ps (design D), and 66.10 ps (design B). Multi-phase average delay for design D is 25.99 ps. Output Q1 rise and fall times are 134.6 ps and 102.6 ps.

Supply current. Design D average supply current is 0.54144 mA (0.541337 mA integrated from transient, matching within 0.018%). Peak current is 0.560 mA (Figure 15).

Comparison with reference literature. Reported values from Kuo et al. (TSMC 180 nm, 1.8 V, 1.4 GHz) are 2.34 mW, phase error within 3.75 deg, and core area 91.7 x 52.9 um^2. In 45 nm, design D at 1.4 GHz consumes 0.5413 mW with 38.46 deg phase error; design B reaches 3.78 deg at 6 GHz.

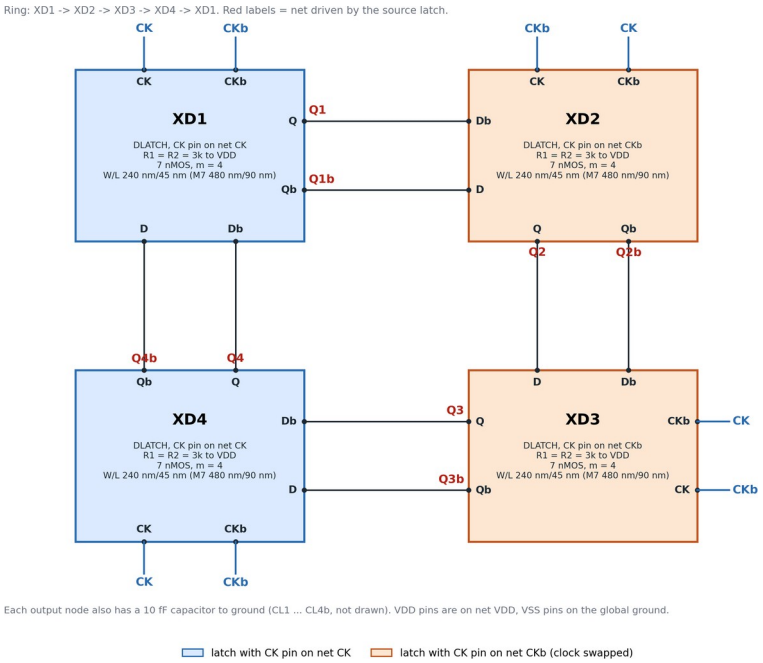


Figure 8: Ring interconnection of four DLATCH stages (XD1 to XD4) with alternating cross-connections, and inverted clocks on XD2 and XD3 (design D: m = 4, R = 3 kOhm).

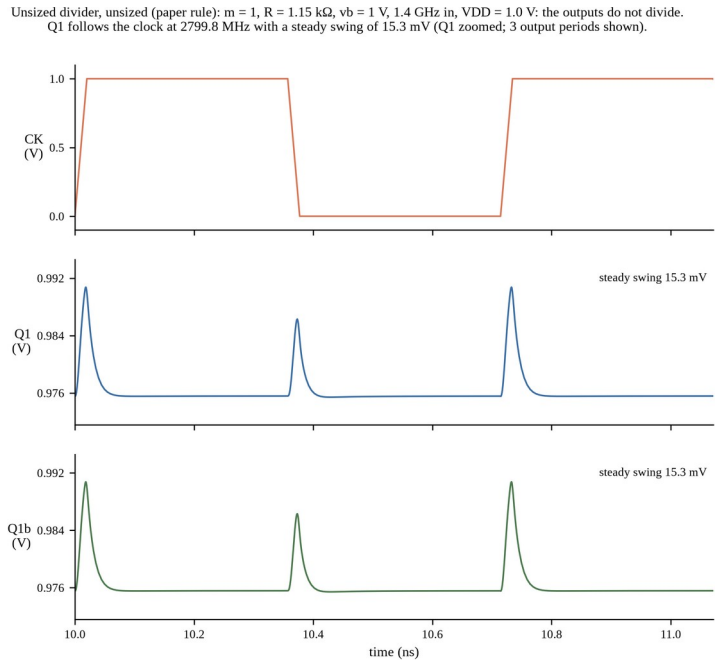


Figure 9: Transient simulation with paper sizing (m = 1, R1 = R2 = 1.15 kOhm = Ron/2) at 1.4 GHz. Division fails: Q1 oscillates at 2799.8 MHz with only 15.3 mV swing (0.975 V to 0.991 V).

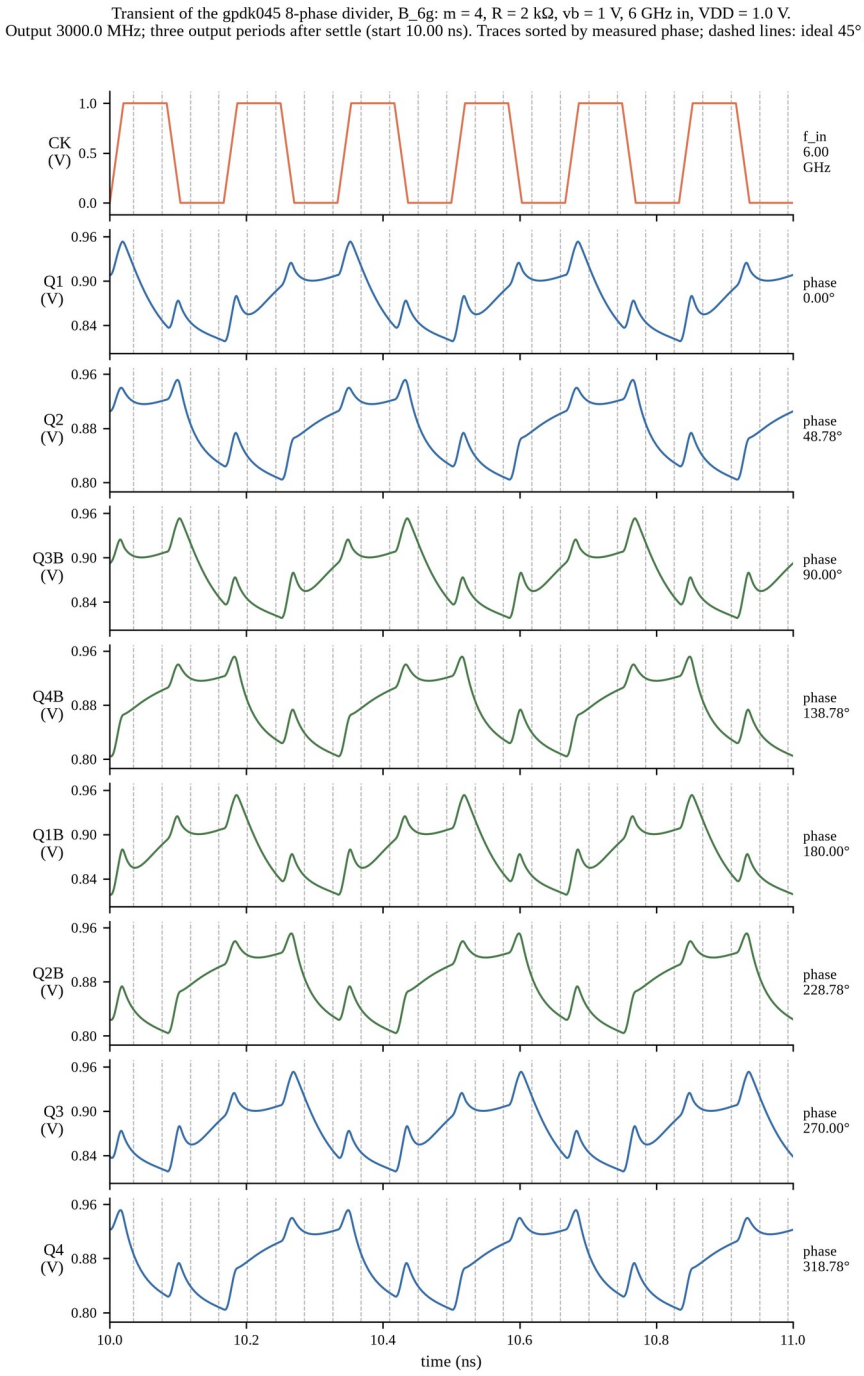


Figure 10: Transient waveforms for design B at 6 GHz (m = 4, R1 = R2 = 2 kOhm), yielding divided outputs at 3000.0 MHz.

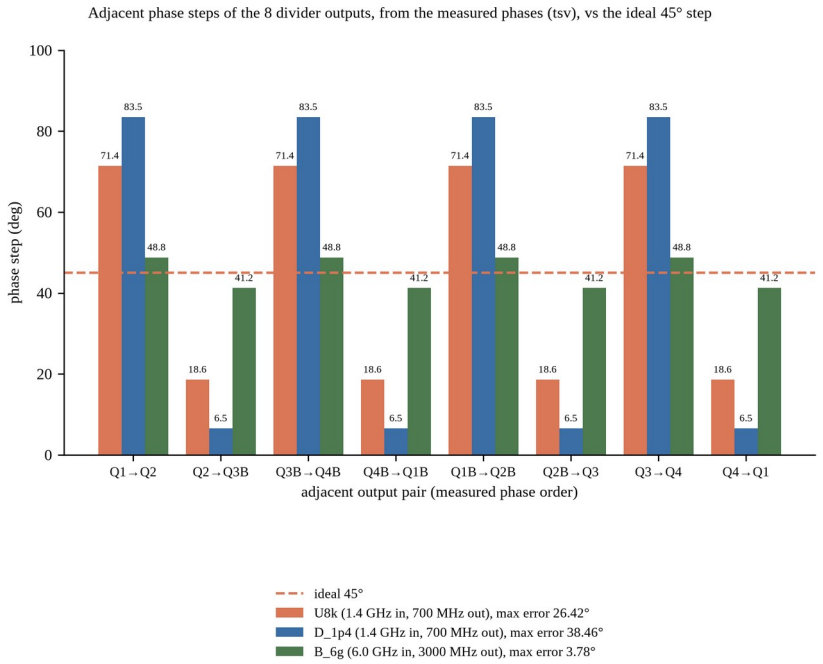


Figure 11: Phase steps between the eight outputs for unsized baseline (1.4 GHz), design D (1.4 GHz), and design B (6 GHz).

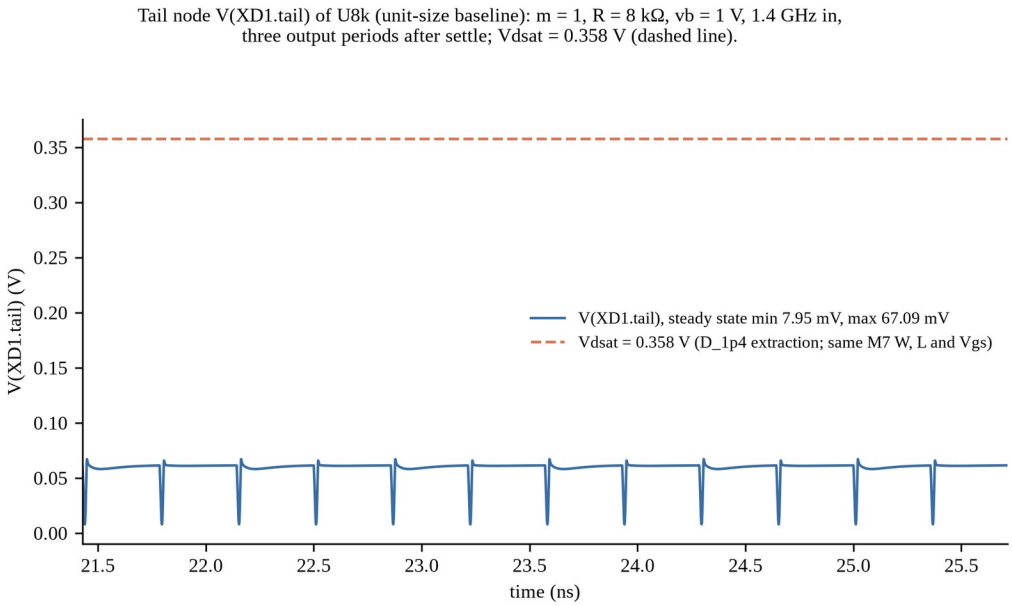


Figure 12: Tail transistor (M7) drain voltage for unsized baseline compared with saturation voltage Vdsat = 0.358 V.

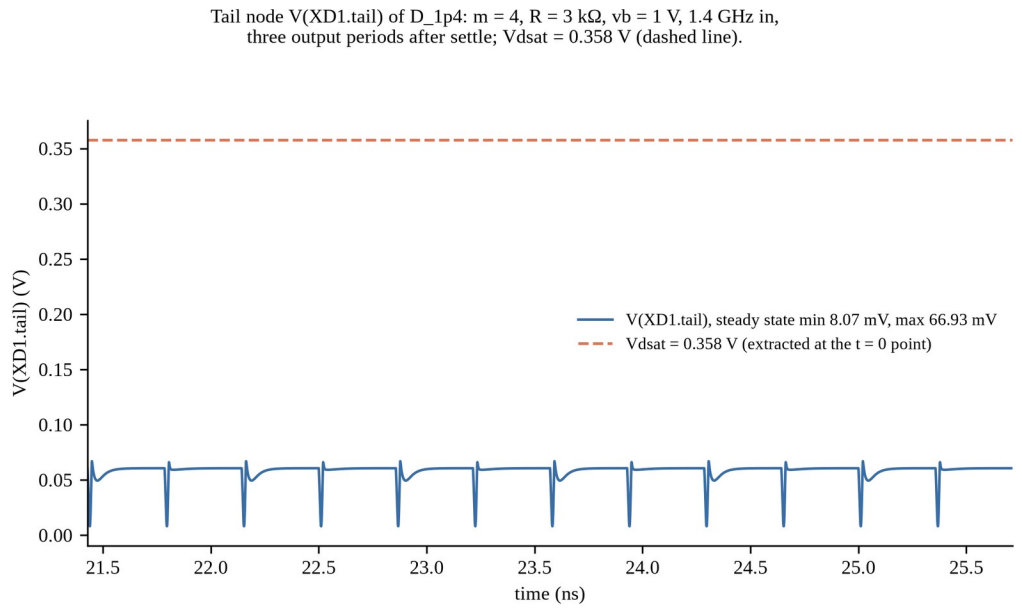


Figure 13: Tail transistor (M7) drain voltage for design D at 1.4 GHz compared with saturation voltage Vdsat = 0.358 V.

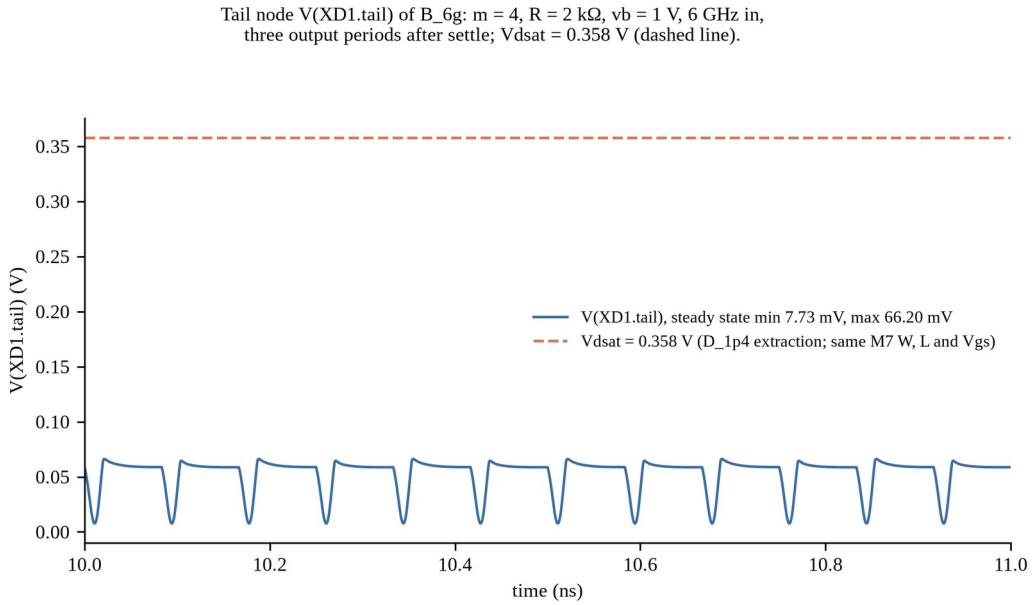


Figure 14: Tail transistor (M7) drain voltage for design B at 6 GHz compared with saturation voltage Vdsat = 0.358 V.

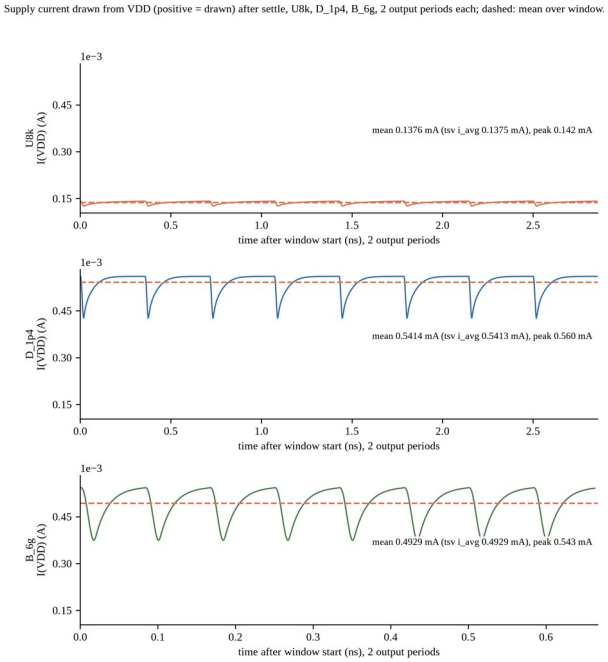


Figure 15: Supply current drawn from VDD for unsized baseline, design D, and design B.

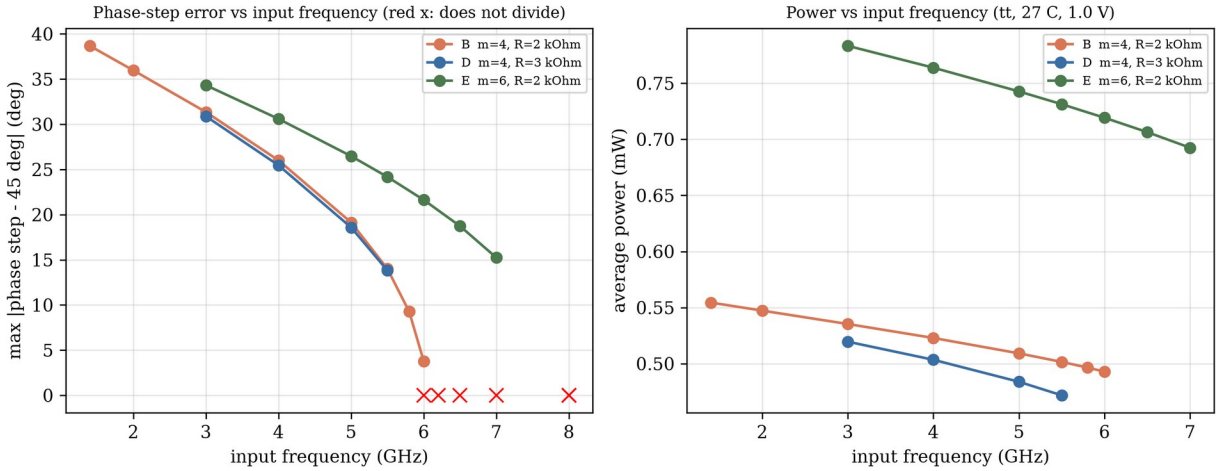


Figure 16: Frequency sweeps for designs B, D, and E showing division range and maximum phase-step error versus input frequency.

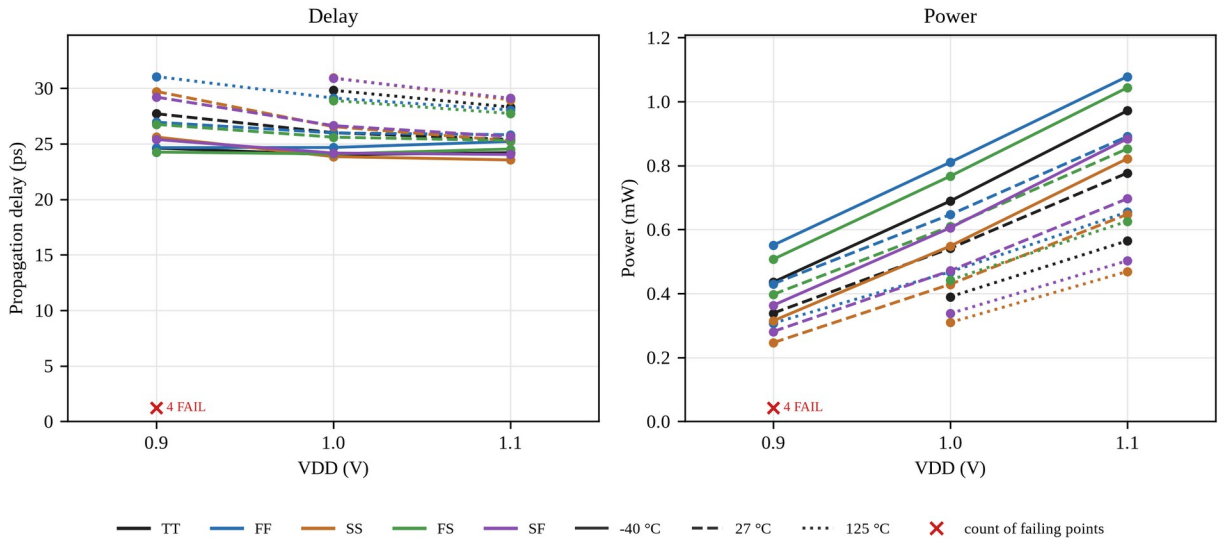


Figure 17: Delay and power across the PVT matrix for design D at 1.4 GHz (crosses mark failing corners).

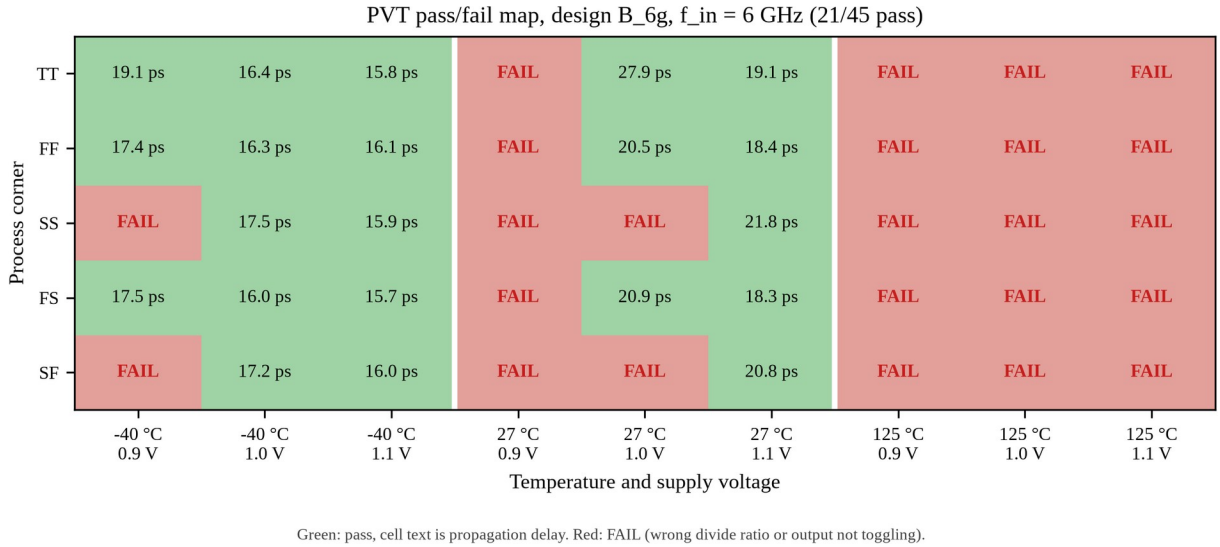


Figure 18: PVT pass/fail map for design B at 6 GHz (21 of 45 points pass).

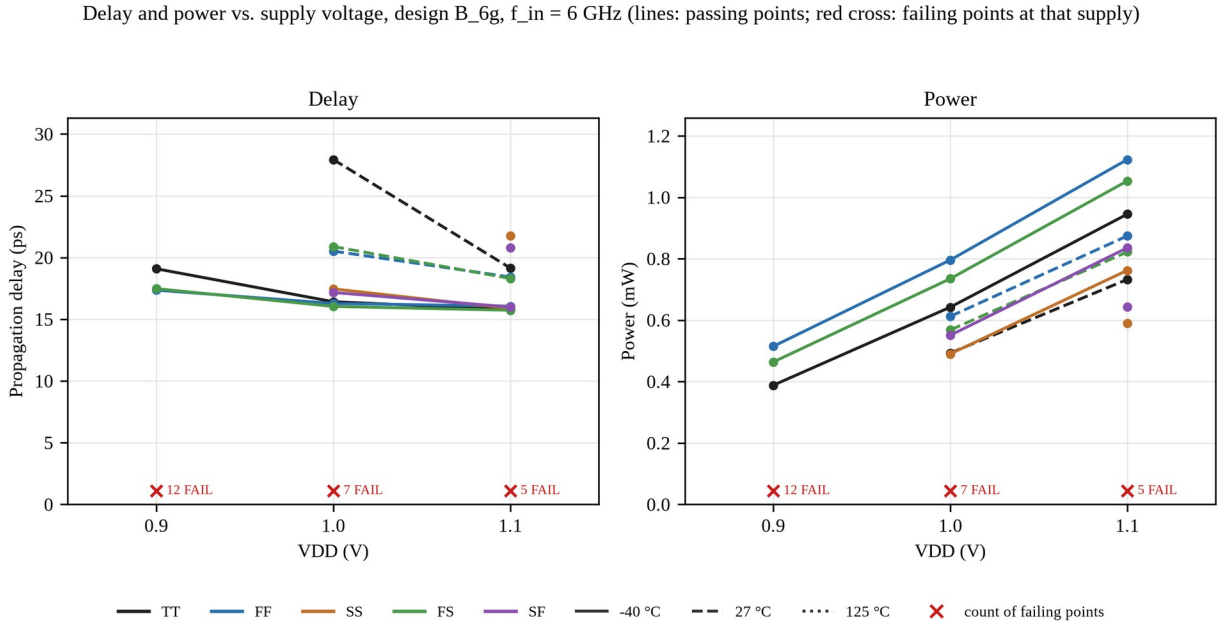


Figure 19: Delay and power across the PVT matrix for design B at 6 GHz (crosses mark failing corners).

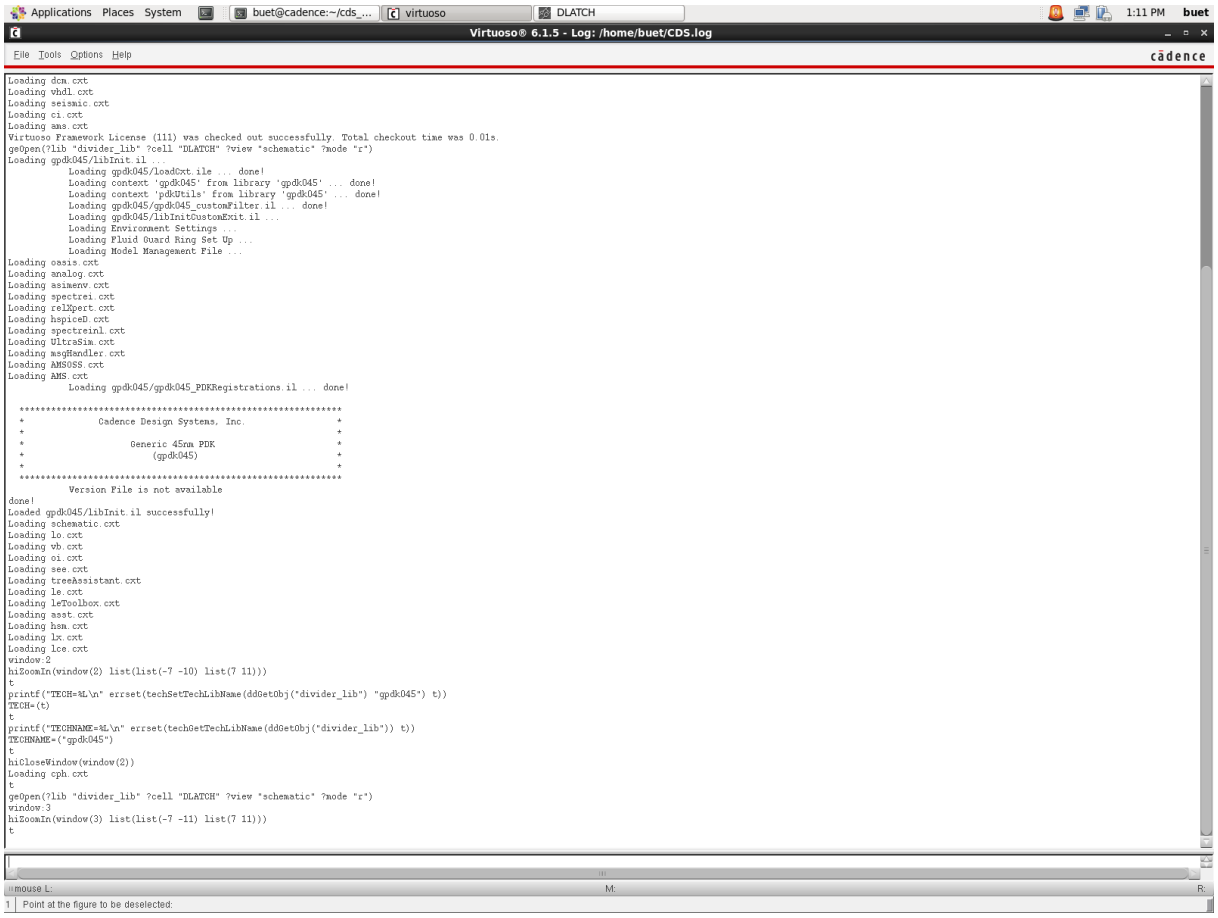


Figure 20: Virtuoso Command Interpreter Window (CIW) log confirming gpdk045 library initialization.

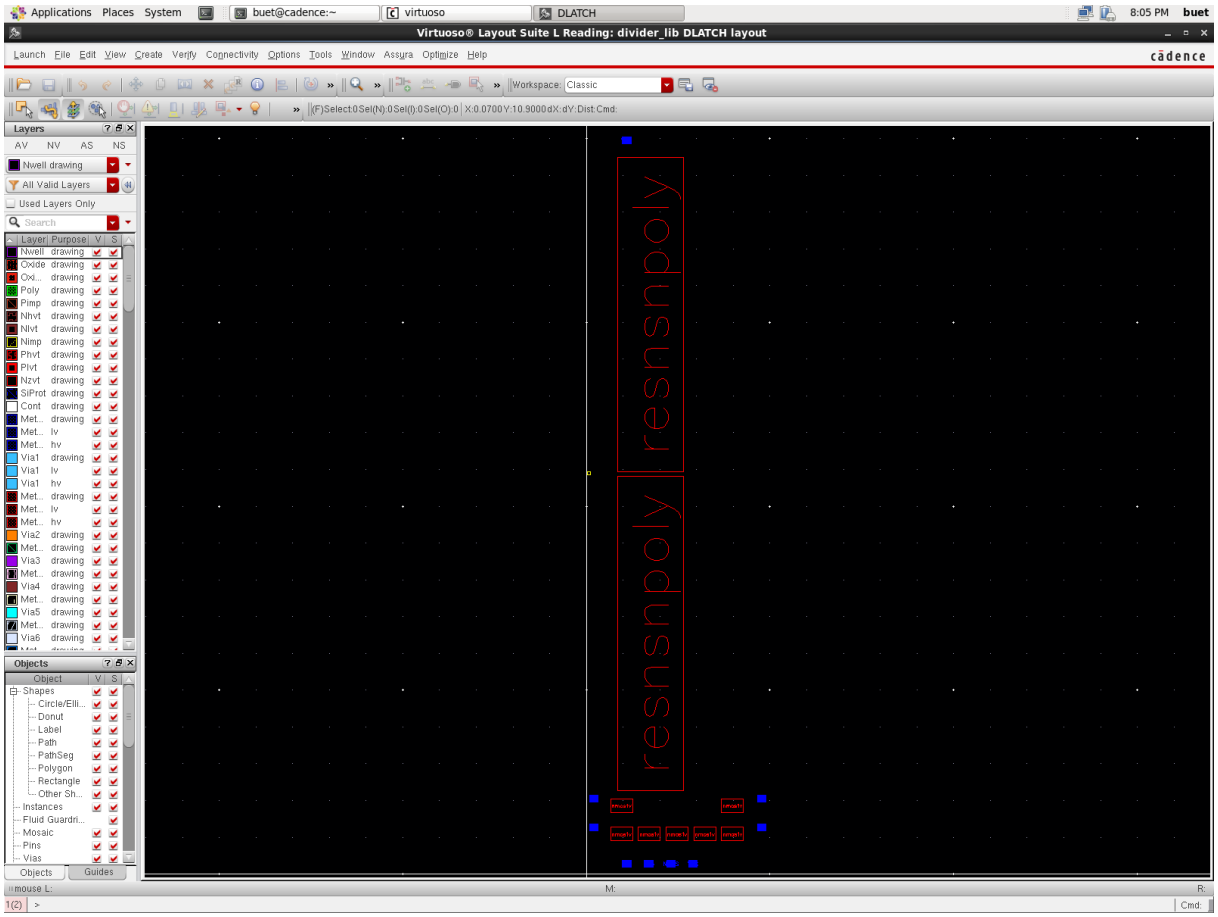


Figure 21: Virtuoso layout editor showing the DLATCH cell: 7 unit nMOS devices (m = 1) and 2 poly resistors (placement only, unrouted).

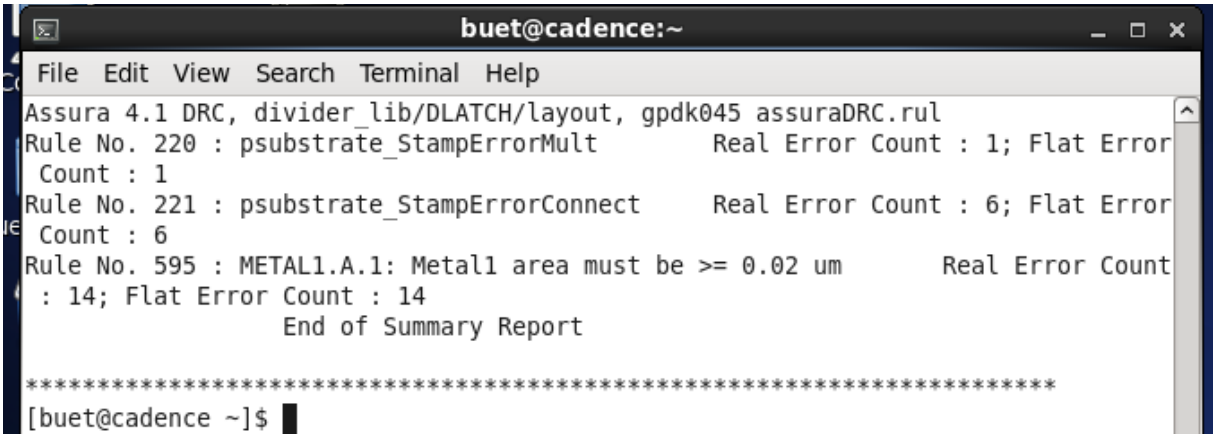


Figure 22: Assura DRC summary for the DLATCH layout showing 21 errors across 3 rules.

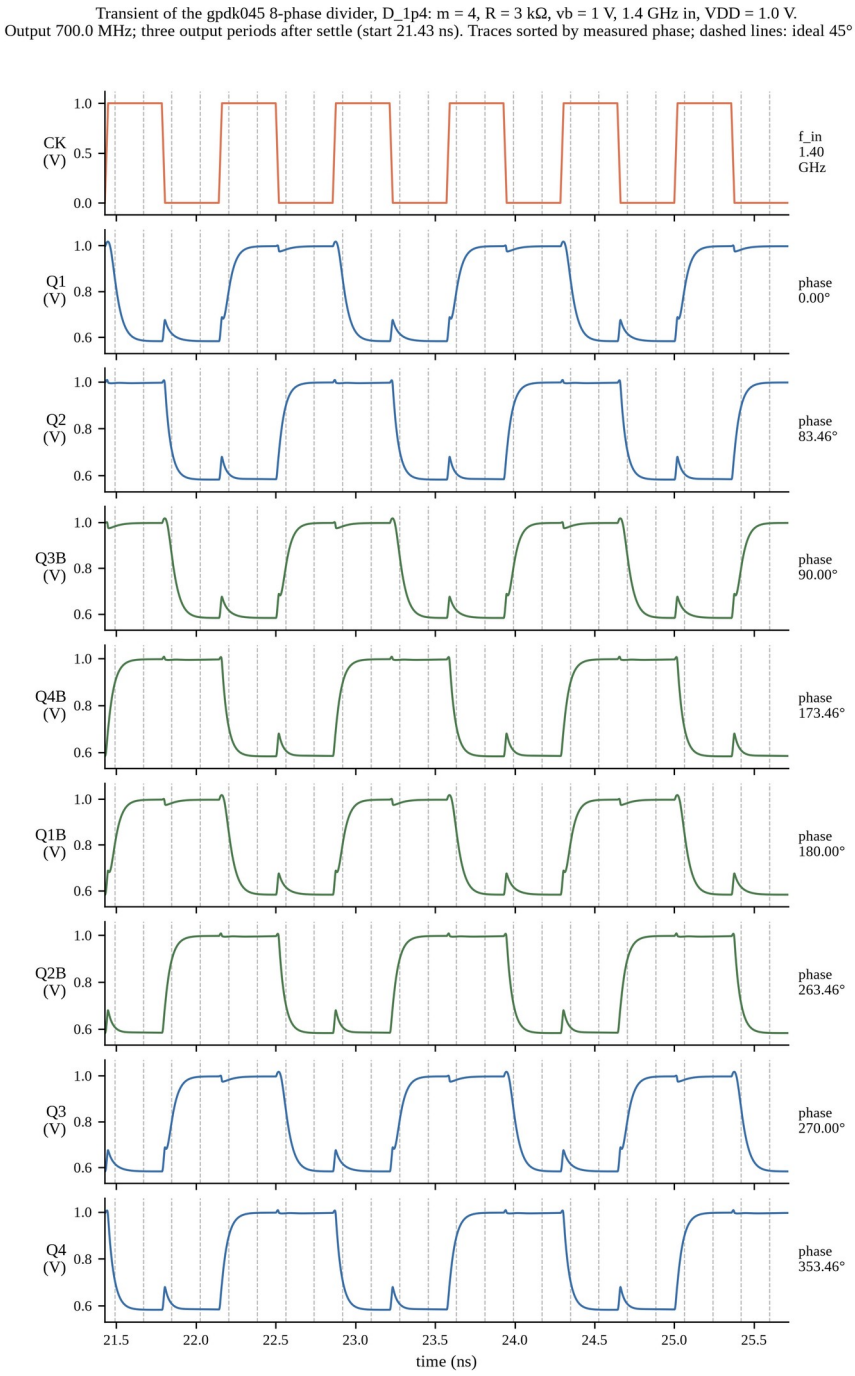


Figure 23: Transient waveforms for design D (m = 4, R1 = R2 = 3 kOhm) at 1.4 GHz. Output frequency is 700.0 MHz with Q1 swinging from 0.583 V to 1.017 V. Tail voltage V(XD1.tail) remains between 8.1 mV and 66.9 mV (XD2 reaches 68.2 mV).

CONCLUSION:

Multiphase CMOS division in 45 nm was demonstrated at two operating points using the stack resistance rule. Design D at 1.4 GHz (m = 4, R = 3 kOhm) achieved 0.5413 mW, 25.99 ps delay, 14.067 fJ PDP, and 38.46 deg phase error (41 of 45 PVT corners passing). Design B at 6 GHz (m

= 4, R = 2 kOhm) achieved 0.4929 mW, 27.94 ps delay, 13.768 fJ PDP, and 3.78 deg phase error (21 of 45 PVT corners passing). Design B exhibited reduced swing (0.135 V, 0.818 V to 0.953 V against the 0.1 V limit), failed at tt, 27 C, 0.9 V, and failed division at 6.2 GHz.

The paper sizing rule (m = 1, R1 = Ron/2 = 1.15 kOhm) fails to divide at 1.4 GHz because load resistance is below the regeneration threshold (7.59 kOhm). Empirical division thresholds span 7 to 8 kOhm at vb = 0.8 V, and 6 to 7 kOhm at vb = 1 V.

Operating limitations: transistor M7 operates in triode across all designs. Phase-step error at 1.4 GHz (38.46 deg) exceeds the literature value (3.75 deg); 3.78 deg was reached only at 6 GHz. All 4 PVT failures for design D occur at 125 C and 0.9 V. Design B operates near the regeneration boundary, passing 21 of 45 PVT points at 6 GHz and 27 of 45 at 1.4 GHz.

Layout implementation: placement was performed for unit-size transistors (m = 1) and placeholder resistors (2000 ohm); routing, LVS, extraction, and post-layout simulation were not performed (design D with m = 4 and R = 3 kOhm was not laid out). DLATCH DRC yields 21 errors across 3 rules (7 substrate connectivity, 14 Metal1 minimum area).

PROBLEMS FACED DURING THE IMPLEMENTATION:

The paper sizing rule does not divide at unit device size because the load resistance falls below the regeneration threshold.

Conventional division checks initially counted false passes caused by low-amplitude harmonics at 2 f_in.

Tail current source transistor M7 operates in triode rather than saturation across all operating conditions.

Spectre BSIM4 model compilation failed on the 32-bit simulation host due to missing compiler and runtime libraries.

Layout automation (generate-from-source and automatic router) completed 136 of 137 connections but produced 134 DRC errors.

PROBLEMS RESOLVED USING SOLUTIONS:

The load resistor was resized using the 3 Ron/m stack resistance rule, establishing R = 8 kOhm for the baseline and R = 3 kOhm for design D.

A strict eight-output verification check was introduced requiring division to f_in/2 within 0.1% and at least 0.1 V peak-to-peak swing.

Tail node transient voltages were characterized against extracted BSIM4 Vdsat, confirming resistive tail degeneration mode.

Spectre model compile problems on the 32-bit simulation host were resolved by installing the missing compiler and runtime libraries.

The placement layout without routing was reported with a full tally and characterization of the 21 Assura DRC rule violations.

ANY OTHER COMMENT:

Waveform and PVT figures present transient simulation data extracted from Spectre.

On-resistance Ron and division boundaries reflect measured device operating points.

Circuit diagrams reflect netlist connectivity and device parameters for design D.

Reference specifications reflect published literature data (Kuo et al.).

REFERENCES:

Yue-Fang Kuo, Shih-Hung Lin, Bo-Yi Chen and Liang-Chen Liu, "A Compact Multiphase Divide-by-Two CMOS Frequency Divider," IEEE Microw. Wireless Technol. Lett., vol. 35, no. 5, pp. 617-620, May 2025, doi: 10.1109/LMWT.2025.3528863.

Cadence Design Systems, Cadence GPDK045 v6.0 documentation (45 nm generic process design kit).

Cadence Design Systems, Spectre Circuit Simulator Reference, MMSIM 12.1.

APPENDIX: COMPLETE PVT DATA

Complete 45-corner simulation results for design D at 1.4 GHz across process corners (tt, ff, ss, fs, sf), temperatures (-40, 27, 125 °C), and supply voltages (0.9, 1.0, 1.1 V). Total 41 corners pass division to 700.0 MHz; 4 corners fail (all at 125 °C, 0.9 V for tt, ss, fs, sf).

Table 6: Complete PVT analysis results for design D at 1.4 GHz (45 corners)

Corner	Temp (°C)	VDD (V)	Divides	f_out (MHz)	Delay (ps)	Power (mW)	PDP (fJ)	Note
tt	-40	0.9	PASS	700.0	24.61	0.4356	10.72	
tt	-40	1.0	PASS	700.0	24.01	0.6892	16.54	
tt	-40	1.1	PASS	700.0	24.19	0.9714	23.50	
tt	27	0.9	PASS	700.0	27.70	0.3387	9.38	
tt	27	1.0	PASS	700.0	25.99	0.5413	14.07	
tt	27	1.1	PASS	700.0	25.38	0.7769	19.72	
tt	125	0.9	**FAIL**	2799.7	n/a	n/a	n/a	outputs at 2 f_in, swing 0.034 V < 0.1 V
tt	125	1.0	PASS	700.0	29.81	0.3898	11.62	
tt	125	1.1	PASS	700.0	28.30	0.5648	15.98	
ff	-40	0.9	PASS	700.0	24.65	0.5512	13.59	
ff	-40	1.0	PASS	700.0	24.66	0.8104	19.99	
ff	-40	1.1	PASS	700.0	25.20	1.0784	27.17	
ff	27	0.9	PASS	700.0	26.94	0.4303	11.59	
ff	27	1.0	PASS	700.0	25.97	0.6467	16.80	
ff	27	1.1	PASS	700.0	25.81	0.8910	23.00	
ff	125	0.9	PASS	700.0	31.05	0.3074	9.55	
ff	125	1.0	PASS	700.0	29.11	0.4677	13.61	
ff	125	1.1	PASS	700.0	28.04	0.6547	18.36	
ss	-40	0.9	PASS	700.0	25.60	0.3147	8.06	
ss	-40	1.0	PASS	700.0	23.84	0.5483	13.07	
ss	-40	1.1	PASS	700.0	23.54	0.8217	19.34	
ss	27	0.9	PASS	700.0	29.71	0.2463	7.32	
ss	27	1.0	PASS	700.0	26.54	0.4288	11.38	

Corner	Temp (°C)	VDD (V)	Divides	f_out (MHz)	Delay (ps)	Power (mW)	PDP (fJ)	Note
ss	27	1.1	PASS	700.0	25.27	0.6476	16.37	
ss	125	0.9	**FAIL**	2799.7	n/a	n/a	n/a	outputs at 2 f_in, swing 0.029 V < 0.1 V
ss	125	1.0	PASS	700.0	30.93	0.3095	9.57	
ss	125	1.1	PASS	700.0	28.94	0.4688	13.57	
fs	-40	0.9	PASS	700.0	24.25	0.5072	12.30	
fs	-40	1.0	PASS	700.0	24.08	0.7669	18.46	
fs	-40	1.1	PASS	700.0	24.52	1.0436	25.59	
fs	27	0.9	PASS	700.0	26.74	0.3971	10.62	
fs	27	1.0	PASS	700.0	25.59	0.6098	15.61	
fs	27	1.1	PASS	700.0	25.29	0.8525	21.56	
fs	125	0.9	**FAIL**	700.0	n/a	n/a	n/a	divides, swing 0.093 V < 0.1 V
fs	125	1.0	PASS	700.0	28.89	0.4423	12.78	
fs	125	1.1	PASS	700.0	27.72	0.6259	17.35	
sf	-40	0.9	PASS	700.0	25.38	0.3632	9.22	
sf	-40	1.0	PASS	700.0	24.16	0.6055	14.63	
sf	-40	1.1	PASS	700.0	24.03	0.8837	21.23	
sf	27	0.9	PASS	700.0	29.20	0.2811	8.21	
sf	27	1.0	PASS	700.0	26.63	0.4713	12.55	
sf	27	1.1	PASS	700.0	25.65	0.6964	17.86	
sf	125	0.9	**FAIL**	2799.7	n/a	n/a	n/a	outputs at 2 f_in, swing 0.031 V < 0.1 V
sf	125	1.0	PASS	700.0	30.89	0.3375	10.43	
sf	125	1.1	PASS	700.0	29.09	0.5023	14.61	